SharpeBench: The Luck-Robust Benchmark for AI Trading Agents

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Abstract

Benchmarks for AI trading agents rank on realized return over short windows, which measures the variance of noise more than skill: one widely cited board places its leader at a Sharpe of 1.51 ± 1.08 and its buy-and-hold baseline at 0.02 ± 0.87, intervals that overlap. We describe SharpeBench, a deterministic, judge-free benchmark that ranks an agent only if its edge survives deflation for the number of strategies tried, reliability across every execution seed and out-of-sample window, a process audit that zeroes any entry earned by bypassing a risk control, and a block-bootstrap null. Raw return never ranks. We run it across nine frozen datasets spanning four asset classes and four bar sizes and report four findings. The deflation prior the benchmark shipped with was annualized while the kernel computes Sharpe per period, so a daily agent needed an annualized Sharpe of 18 to rank and an hourly one 106; we correct the protocol. With units corrected, no reference agent is eligible anywhere, because every dataset contains a bear window and the reliability gate requires profitability in every window: the benchmark declines to certify that owning the index is safe in a downturn: the market itself cannot pass it. A weaker gate bounding per-window drawdown refuses the same agents, which drew down between 32 and 99 percent in their worst windows. And a risk-managed agent with no alpha, added to test whether eligibility is vacuous, is refused solely on deflation after clearing every other gate: the gates discriminate, demoting beta on reliability and discipline-without-edge on deflation. The luck floor of random agents never beats a reference agent, and with deflation off a random agent reaches a deflated Sharpe of 0.999. The benchmark ships with an eight-attack self-audit, replay-recompute tamper evidence, cross-platform golden fixtures, a publicly verifiable signed board, and a forward arena that fixes scoring rules before data exists and chains its published boards into one verifiable history. Every number is produced by a listed command from committed data. External validation against live performance, and the alpha-bearing agent that would pass, are the outstanding experiments.

1 Introduction

Give a thousand random agents a quarter of market data and one of them will look like Renaissance. Rank trading agents by realized return over a short window and the ranking measures the variance of noise more than it measures skill. Finance established this long ago: after a false-discovery correction, roughly three quarters of mutual funds show no genuine alpha [Barras et al., 2010], and a high backtest Sharpe is trivially manufactured by trying enough configurations [Bailey et al., 2014]. The benchmarks now used to rank AI trading agents inherit none of that caution. One ranks on a

Sharpe ratio whose own confidence interval spans from 0.43 to 2.59 [Xie et al., 2024]. Another evaluates on a single four-month window with no deflation [Chen et al., 2025]. A 2026 evidence map of LLM-trading research coded its nineteen primary empirical studies for reproducibility and found that none reached the top tier [Xia et al., 2026]. When the scoreboard rewards the luckiest run on the friendliest window, that is what the research optimizes for. For a coding agent a flattering benchmark wastes some time. For an agent that allocates capital it selects the strategy most likely to blow up, the one whose backtest looked best because it overfit hardest.

This paper describes SharpeBench, a benchmark that ranks trading agents not on how much they made but on whether their edge is real, and reports what happened when the benchmark was applied to the market itself. The construct it measures is skill that survives four corrections the literature already supplies and no existing agent board applies as gates: deflation for the number of strategies tried [Bailey and López de Prado, 2014], reliability across every execution seed and out-of-sample window [Yao et al., 2024], a process audit that zeroes any entry earned by bypassing a risk control, and a block-bootstrap null. An agent ranks only if all four hold. Raw return is reported and never ranks. The scorer is deterministic, judge-free, and byte-identical on any platform, and it ships with a self-audit that fires eight known attacks at itself on every commit.

The paper's evidence is the benchmark run across nine frozen datasets spanning four asset classes and four bar sizes, from hourly crypto to weekly equity indices, with a field of reference agents and a luck floor of random agents on each. Four findings came out of it, and two of them are corrections to the benchmark as it shipped.

- **The deflation prior was in the wrong units.** The literature supplies the dispersion of Sharpe ratios across trials annualized; the kernel computes Sharpe per period. Applied per period, the default prior demanded an annualized Sharpe of 18 on daily bars and 106 on hourly bars before any agent could rank. The market is near 0.6. The error grows with the square root of the number of periods per year, which is why the same index scored 0.984 on weekly bars and 0.000 on daily ones. Section 5.2 quantifies it and Section 3.1 states the corrected protocol.
- **On every real dataset the reliability gate is the one that bites.** With units corrected, every dataset still admits no agent, and the reason is that every dataset contains a bear window and the default reliability mode requires profitability in every window. Buy-and-hold is structurally ineligible on any dataset containing a downturn. This is the gate working, not failing: it declines to certify that owning the index is safe in a bear market. Section 5.3 shows the two gates separating cleanly in the data.
- **A weaker reliability gate admits nobody either.** An ablation with a per-run drawdown bound in place of every-regime profitability, made expressible from configuration for this purpose, refuses the same agents, because the reference agents drew down between 32 and 99 percent in their worst windows. A weekly crypto momentum agent that clears the deflation bar at 0.978 is refused on a 56 percent drawdown. A raw-return leaderboard ranks it first. Section 5.4 reports the ablation.
- **Discipline without alpha is refused on deflation alone.** A risk-managed agent with a trend filter, volatility targeting and a drawdown halt, added with documented defaults and no tuning, clears the bootstrap, the probabilistic Sharpe, the process gate and the per-run drawdown bound on weekly US indices and is refused solely on a deflated Sharpe of 0.30 against the 0.95 bar. The gates discriminate rather than blanket-refuse: beta fails reliability, discipline without edge fails deflation. Section 5.5 reports it, including the two datasets where the discipline itself whipsaws.

Contributions. (1) A luck-robust, reliability-gated, process-gated, deterministic and forward-attested benchmark for trading agents, with the exact eligibility predicate stated in Equation (3). (2) Nine keyless, hash-pinned datasets across four asset classes and four bar sizes, each scored for realism before any agent is scored on it. (3) Four empirical findings from running the benchmark on the market, two of which corrected the benchmark and one of which shows the gates discriminate. (4) A self-audit of eight attacks, a replay-recompute tamper check, a publicly verifiable signed board, and a forward arena whose published boards chain into one verifiable history, all exercised by tests on every commit. (5) An import adapter that turns re-scoring a rival board's field under these gates into one command. Every number in the paper is produced by a command in Section A.

2 Design principles

Every decision in the benchmark traces to one of seven principles, stated here so the rest of the paper reads as their consequences. **Luck-robust**: a score that does not deflate for the number of strategies tried is a score of the best overfitter, so the deflated Sharpe is a gate and each agent's declared trials raise its own bar. **Reliability-gated**: an agent that is safe on average is not safe, so eligibility requires clearing the per-run bar on every seed and every window, which is a stronger claim than having an edge and is meant to be. **Process-gated**: a return earned by bypassing a risk control is not a return, and one block-severity violation in any run zeroes the entry. **Point-in-time**: the agent interface can express only history at or before the decision, and that boundary is the thing to audit. **Deterministic**: a score is a pure function of a submission and a configuration, byte-identical on any platform, and the continuous-integration matrix checks that rather than assumes it. **Forward-attested**: trust comes from a commitment made before the data exists and a chain anyone can verify, not from the host's word, and the host competes on its own board. **Judge-free**: no model grades another, every gate is an assertion over numbers, and that is what makes the self-audit a property of the code rather than an opinion about it.

Two consequences are worth stating before the experiments, because the experiments are about them. The first principle requires a dispersion of Sharpe ratios across trials, which the literature supplies annualized while the kernel computes Sharpe per period; the fifth makes the mismatch visible, since a reproducible scorer is wrong in exactly the same way everywhere. The second and third together mean the benchmark can decline to certify any agent at all on a dataset, including the market itself. That is not a failure of the benchmark. Whether it is the right outcome is a question the paper puts to the data.

3 The benchmark

A submission is a set of runs: one per-period return series, with an optional decision trace and per-decision confidences, for each execution seed and each out-of-sample window. The scorer turns a field of submissions into a ranked board. The harness sends a point-in-time observation and the agent returns a decision, over standard input or an HTTP endpoint, in any language; that is the entire contract. This section states what the scorer computes, in the units it computes it, and which statistics decide rank.

3.1 Statistics

Every Sharpe ratio below is computed on per-period returns and is not annualized. The distinction is the subject of the first finding in Section 5, so it is stated once here and relied on throughout.

Probabilistic and deflated Sharpe. For an observed per-period Sharpe $\widehat{SR}$ over a track of length n with sample skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$, the probability that the true Sharpe exceeds a benchmark SR^* is [Bailey and López de Prado, 2014]

$$\widehat{PSR}(SR^*) = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{n-1}}{\sqrt{1 - \hat{\gamma}_3\,\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\,\widehat{SR}^2}}\right). \tag{1}$$

Fat tails and negative skew lower it for the same headline Sharpe. The deflated Sharpe is the PSR evaluated against the Sharpe one expects from the best of N trials with dispersion σ_{trials},

$$SR_0^* = \sigma_{\text{trials}}\left[(1-\gamma)\,Z^{-1}\big(1 - \tfrac{1}{N}\big) + \gamma\,Z^{-1}\big(1 - \tfrac{1}{Ne}\big)\right], \tag{2}$$

with γ the Euler–Mascheroni constant and Z^{-1} the normal quantile. N is the field size plus each agent's declared in-sample trials, so a strategy mined from a thousand private backtests is deflated for

Table 1: Frozen datasets. Bars are per symbol. Periods per year is what every annualized threshold is converted with.

Dataset	Symbols	Bars	Periods/yr	Realism
US indices 1d	SPX DJI IXIC	7540	252	pass
US indices 1w	same	522	52	fail, short
Crypto majors 1h	BTC ETH SOL BNB XRP	24000	8760	pass
Crypto majors 4h	same	6000	2190	pass
Crypto majors 1d	same	5001	365	pass
Crypto majors 1w	same	307	52	pass
FX majors 1d	EUR GBP JPY AUD CHF vs USD	4157	252	fail, Zumbach
Commodities 1d	WTI Brent	4126	252	pass
Rates 1d	10y Treasury yield	4157	252	pass

that search. σ_{trials} is annualized in the literature it comes from and per period inside Equation (2); the scorer takes it annualized and divides by $\sqrt{\text{periods per year}}$ once at the point of use. When the field holds at least five agents it instead measures the dispersion of per-period Sharpes across the field, which needs no conversion, and records which path was taken on every score.

Reliability, significance, process. Following pass^k [Yao et al., 2024], each run passes if its own PSR clears a per-run bar and the agent passes under a mode; the default requires every run across every seed and window, because an agent that is safe on average is not safe. A stationary block bootstrap [Politis and Romano, 1994] with a fixed seed tests each agent against a null of no edge while preserving serial correlation. The field-wide Reality Check [White, 2000], superior-predictive-ability test [Hansen, 2005] and Romano–Wolf step-down [Romano and Wolf, 2005] are computed and reported on every row; they do not gate, and earlier documentation that called them gates was wrong. Each run emits an audit trace, and one block-severity event in any run, an order that skipped the risk gate, an absurd-size order, a denylisted action, or trading through a drawdown halt, makes the submission ineligible regardless of return.

3.2 The eligibility predicate

An agent is rank-eligible if and only if

$$\text{DSR} \geq \beta \ \wedge \ \text{pass}^k \ \wedge \ \text{process} \ \wedge \ p_{\text{boot}} < \alpha \ \wedge \ \text{mandate}, \tag{3}$$

with defaults $\beta = 0.95$, $\alpha = 0.05$, a per-run PSR bar of 0.90, and a mandate bounding drawdown over the pooled track and over any single run. Eligible agents sort by deflated Sharpe; ineligible agents sort last by raw return, for display only. Raw return never buys rank. Everything else the scorer computes, alpha and beta against the field, calibration, the half-life of the information coefficient, the three field-wide tests, drawdown, turnover, Sortino and a rolling worst-window Sharpe, is reported beside the rank and does not move it, so a high score is legible because its reasons are on the same row.

3.3 Simulator and data

Execution charges fees, seeded slippage, square-root own-order impact and financing under three named cost profiles; all experiments use the typical profile, and the seeded slippage is what gives pass^k different runs to measure. Nine keyless, hash-pinned datasets ship across four asset classes and four bar sizes (Table 7), and each is scored against the stylized facts of Cont [2001] before any agent is scored on it. Seven pass. Weekly US indices fail the aggregation test because 522 bars is too short for it, and FX majors fail time-reversal asymmetry, which is weak in that market; both ship with the verdict recorded, because a realism gate that never fails is not a gate. Single-name equities are absent for want of a keyless feed. Section C gives the cost parameters and the data caveats.

Table 2: The self-audit battery. Each attack beats the honest reference agent on raw return and is demoted by the gate named. All eight are demoted on every commit.

Attack	What it exploits	Demoted by
luck-not-skill	one lucky seed, highest raw return	pass^k
risk-gate-bypass	an order that skipped the risk gate	process
sim-exploitation	an absurd-size order	process
mandate-breach	exceeding the drawdown mandate for return	mandate
raw-return-cannot-buy-rank	top raw return on some runs only	pass^k
cheat-reward-hacker	gate bypass plus padded confidence	process
tail-seller	smooth returns from selling tail risk	process
adversarial-input	accurate forecasts, collapse under perturbation	pass^k

4 Threats to validity and the integrity protocol

A benchmark that scores capital allocation is a target. This section lists the ways it could be gamed or could mislead, and for each says what the benchmark does and how a reader can check.

Look-ahead. The observation builder hands the agent the trailing window of history at or before the decision index, and the agent never holds a handle to the dataset. A future bar is therefore unrepresentable through the agent interface. This is a narrower claim than earlier documentation made: the dataset type exposes a close-at-index accessor that accepts any index, so look-ahead is prevented at the agent boundary, not by the type system. It holds for every shipped transport, and the observation builder is the thing to audit.

Determinism. The kernel performs no input or output, reads no clock, draws no ambient randomness, and sums in a fixed order; every bootstrap takes an explicit seed and every map is an ordered tree. Golden fixtures commit the full ranked output of two fields at full floating-point precision, a one-unit change in the last place of any statistic fails the test, and regeneration is a deliberate local command that refuses to run under continuous integration. The fixtures are checked on Linux, macOS and Windows on every commit, and a parity test asserts that the WebAssembly build and the native kernel produce byte-identical scores. Before this paper the cross-platform claim was architectural and the continuous-integration check compared a run to itself on one machine; both are now checks.

Self-audit. Surveys of benchmark integrity [Reuel et al., 2024, Kapoor et al., 2024] find that benchmarks with a model judge or a single tunable target get gamed. This benchmark has no judge and is deterministic, so its resistance is something to run. `sharpebench audit` constructs eight attacking submissions, scores each against the live kernel, and exits non-zero if any is not demoted; the same battery is a unit test on every commit, so a change that weakens a gate fails the build. Table 2 lists them. The eighth is adapted from Deza et al. [2021], whose finding is that forecast error alone is not sufficient for the trades made on it to lose money: the attacking agent posts excellent returns on the presented series and collapses under a small in-range perturbation of its inputs while its forecast head keeps working, with identical stated confidences and a Brier score of 0.05 against a baseline of 0.25. Its pooled raw return beats the honest agent's. The kernel demotes it through pass^k, and the audit asserts the demotion came from realized returns and not from a process violation, and that calibration could not buy eligibility back. The scorer already defended this case without any change, which is the result a self-audit should produce. Its limit is recorded: a fragility no submitted run exercises is invisible to a kernel with no model, and the harness does not yet generate perturbed windows.

Forward attestation. Every backtest-based score requires the reader to trust that the strategy was not tuned to the window after it was seen. Before a target window's data exists, an entrant publishes a SHA-256 commitment to a digest of its frozen artifact and a salt, revealing nothing; after scoring it reveals both and anyone can verify the match. Published boards are signed chains, each entry signed over the previous, so an altered or reordered row breaks the chain. Two schemes are provided and guarantee different things. The HMAC chain is tamper-evident to holders of the shared key, and because HMAC is symmetric the parties who can verify it are exactly those who can forge it. The Ed25519 chain is publicly verifiable: a board carries its verifying key, anyone holding it can check the chain, and no one without the signing key can forge an entry; Ed25519 is deterministic by construction [Josefsson and Liusvaara, 2017], which is why it fits a reproducible kernel. The benchmark shipped with only the HMAC chain and described it as publicly verifiable. It was not, and this paper corrects the claim. The commitment registry advances by an explicit epoch rather than a wall clock, and the arena now drives it: a file-backed window lifecycle fixes the scoring configuration before entries exist, refuses commitments after the deadline epoch, verifies every reveal against its commitment, and publishes an Ed25519 board per window. Boards chain across windows, each anchoring the previous board's final signature, so the arena history is one verifiable object: the tests prove that an edited payload breaks its own chain and that a wholesale re-signed forgery, which self-verifies, is exposed by the next window's anchor. External agents run in a network-isolated container with bounded resources; absent that isolation the arena refuses rather than falling back to host execution. What remains unbuilt is hosting: no HTTP intake and no scheduler, so the arena is driven by an operator or a cron job, and Section 7 says so.

Replay and contamination. A captured trajectory holds only raw decisions, a window and a seed, and a separate verifier recomputes its score; a test asserts that doubling every order recomputes to different returns, so an artifact cannot lie about its returns. Sealed held-out data under a published hash, a canary token and a masked view that defeats ticker memorization defend against contamination. Both are specified in Section C.

5 Experiments

Every number in this section is produced by the commands in Section A from the committed data and the kernel at the stated version. All runs use the typical cost profile, eight execution seeds, and six disjoint walk-forward windows sized to the dataset, with a warmup of one tenth of the series clamped to between 20 and 60 bars. The field on every dataset is three reference agents (buy-and-hold, cross-sectional momentum, and hold, which never trades) plus a luck floor of five seeded random agents that are fully invested in random weights each period. The luck floor is the zero-skill distribution an edge must clear. The scoring grid is $\beta \in \{0.80, 0.90, 0.95, 0.99\}$, $N \in \{1, 10, 50, 200\}$, and σ_{trials} either measured from the field or pinned to one of three annualized priors, 64 configurations per dataset, 512 records per dataset, 4,608 records in all.

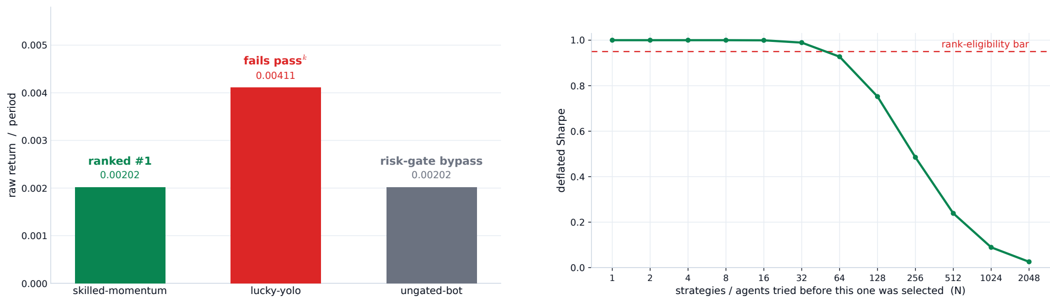
5.1 The demonstration that motivated the benchmark

Figure 1a shows the three-agent field the benchmark shipped with: a skilled momentum agent, a lucky agent with twice its raw return that fails pass^k, and a bot that matched the skilled return but placed one order that never passed its risk gate and is zeroed by the process gate. The agent with the highest raw return ranks last. Figure 1b shows why: one fixed track, scored against a growing pool of trials, crosses the eligibility bar between 32 and 64 trials and collapses past a few hundred. The agent did not change; the context did. Both figures are true and both are computed on synthetic returns. The rest of this section asks what happens on the market.

5.2 Finding 1: the deflation prior was in the wrong units

The first run of the full grid on the two original daily datasets produced zero rank-eligible agents in every one of the 128 cells. Buy-and-hold on US indices posted a PSR of 1.0000 against zero and a bootstrap p of 0.0005, and scored a deflated Sharpe of exactly 0.0000. A PSR of one and a DSR of zero on the same series cannot both be verdicts about the data. They are a verdict about the threshold.

The kernel computes every Sharpe per period. The default $\sigma_{\text{trials}} = 0.5$ is the worked example from Bailey and López de Prado [2014], and it is an annualized dispersion. Substituted per period into



(a) Three agents, one board.

(b) One track, ever more trials.

Figure 1: (a) The shipped demonstration. (b) The same track scored against a growing pool of trials, per-period Sharpe 0.85 over 150 periods, normal moments. Both computed with the kernel's own formulas.

Table 3: The annualized Sharpe an agent had to exceed to reach the deflation benchmark at $N = 50$, for each prior as it was applied (per period) and each bar size. Kernel at v0.2.1. The value measured from a real field was 0.070.

σ_{trials} as applied	SR_0^* per period	1h	4h	1d	1w
0.0700	0.1590	14.9000	7.5000	2.5000	1.1000
0.2000	0.4550	42.6000	21.3000	7.2000	3.3000
0.3500	0.7970	74.6000	37.3000	12.6000	5.7000
0.5000	1.1380	106.5000	53.3000	18.1000	8.2000

Equation (2) at $N = 50$ it sets $SR_0^* = 1.14$ per period. Table 3 converts that bar back into the annualized Sharpe an agent would need to reach it. On daily bars the requirement was an annualized Sharpe of 18; on hourly bars, 106. The US equity index is near 0.6. Renaissance's Medallion fund is commonly cited near 2 to 3. The bar was not high. It was unreachable, and it grew with the square root of the number of periods per year, which is why the same index scores a DSR of 0.984 on weekly bars and 0.000 on daily bars with nothing about the market having changed.

The rival benchmarks in Section 6 report annualized Sharpe without deflating. This benchmark deflated without annualizing. Both are unit errors, in opposite directions, and the second is harder to notice because it produces a benchmark that appears admirably strict. It also explains, completely, why the demonstration in Section 5.1 had only ever been shown on synthetic inputs: those had per-period returns large enough to clear an annualized bar.

The fix, shipped in v0.3.0, leaves the per-period kernel untouched and states the prior annualized with a periods-per-year field that converts it once at the point of use; Section C gives the tests that pin it.

5.3 Finding 2: on every real dataset, the reliability gate is the one that bites

With the units corrected, the full grid was rerun on all nine datasets. Table 4 reports the default configuration on each. The deflated Sharpe column is unchanged from before the fix on every real dataset, and that is expected: each field holds eight agents, so the scorer measures σ_{trials} from the field, and the measured path was already per period. The values it reports are the true verdict of the statistic. On weekly bars the US index and the crypto momentum agent clear $\beta = 0.95$. On daily bars nothing does, because a 50-trial benchmark against a measured per-period dispersion of 0.070 sits at 0.159 per period and the daily index is near 0.036.

Still, zero agents are eligible at any of the 576 cells. The column that explains it is pass^k, and the reason is visible in the regime labels the simulator assigns to each window: every dataset contains at least one bear window. The default reliability mode requires a per-run PSR of 0.90 on every window. A long-only agent has a negative realized Sharpe in a bear window and cannot clear any positive bar

Table 4: Default configuration ($\beta = 0.95$, $N = 50$, σ_{trials} measured) on every dataset, kernel v0.3.0. Worst run DD is the largest drawdown in any single window. No agent is rank-eligible on any dataset under either reliability verdict. The luck floor never beats a reference agent on raw return.

Dataset	Agent	DSR	pass^k	Worst run DD	Every regime	Never catastrophic
US indices 1w	buy-and-hold	0.9840	no	0.3200	no	no
Crypto 1w	momentum	0.9780	no	0.5630	no	no
Crypto 1w	buy-and-hold	0.7450	no	0.6710	no	no
Crypto 1d	buy-and-hold	0.1440	no	0.4760	no	no
US indices 1d	buy-and-hold	0.0000	no	0.3360	no	no
Crypto 4h	buy-and-hold	0.0000	no	0.4790	no	no
Crypto 1h	buy-and-hold	0.0000	no	0.4900	no	no

there. Buy-and-hold, the baseline every agent is meant to beat, is therefore structurally ineligible on every real dataset the benchmark ships. Weekly US indices isolates the two gates cleanly: a DSR of 0.984 clears the deflation bar and the agent fails on pass^k alone.

This is strictness, not breakage. The default mode does not ask whether an agent has an edge. It asks whether the agent is profitable with 90 percent confidence in every regime and under every execution seed. A regime-dependent edge such as equity beta correctly fails that question. The benchmark is declining to certify that owning the index is safe in a bear market, which it is not.

5.4 Finding 3: a weaker reliability gate admits nobody either

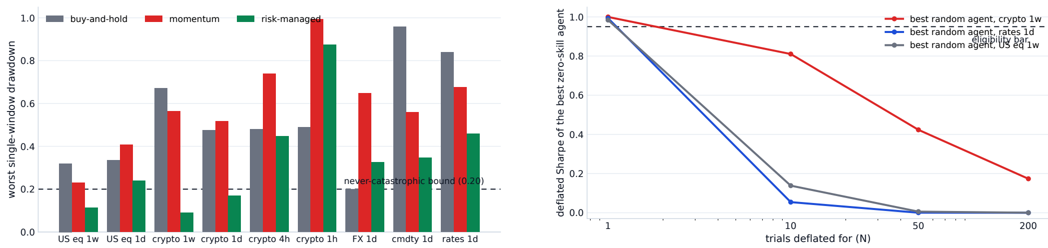
Whether the reliability gate should mean profitable in every regime or never catastrophic in any regime is a design question, and the kernel was extended so that both are expressible from configuration and the ablation is reproducible. The never-catastrophic preset sets the pass mode to any run and adds a per-run drawdown bound of 20 percent, with the edge tested on the pooled track. It is a weaker safety claim, and a test pins its cost: a synthetic agent with one spectacular run and four noise runs is refused by the default only through pass^k and is admitted by the preset.

On the real datasets the preset admits nobody. The last column of Table 4 is identical to the one before it, and the worst-run drawdown column says why. The US index drew down 32 percent in its worst weekly window. Crypto buy-and-hold drew down between 48 and 67 percent, and hourly momentum lost 99.3 percent of its value in a single window, the starkest case for a per-regime bound in the grid. The weekly crypto momentum agent clears the deflation bar at 0.978 and is refused on a 56 percent drawdown in one window. A raw-return leaderboard ranks that agent first. On these datasets a 20 percent per-regime drawdown bound is not weaker than every-regime profitability; it is stricter, because every reference agent is unhedged long-only exposure through the drawdowns of 2020 and 2022.

One correction to how the per-run bound is naturally described. Drawdown is multiplicative, so a run's own drawdown never exceeds the pooled track's, because every within-run peak-to-trough pair is also a pooled pair. A per-run bound at the same value as the pooled bound therefore catches nothing the pooled bound misses. It bites when set below the pooled cap: a loose whole-track budget with a tight per-regime bound. The preset is configured that way, and the test that proves the bound is not redundant constructs exactly that case.

5.5 Finding 4: a disciplined agent without alpha is refused on deflation alone

The findings above show the gates refusing unhedged exposure, which leaves open whether eligibility is attainable at all. To probe that without inventing alpha, a risk-managed reference agent was added to the field: long the equal-weight basket only when it trades above its trailing mean, gross exposure scaled to an inverse-volatility target, and fully flat after a 10 percent drawdown with re-entry only on a fresh trend signal. It is textbook risk management and deliberately nothing more; every parameter is a documented default and none was tuned against the gate.



(a) Worst single-window drawdowns.

(b) Deflation against zero-skill agents.

Figure 2: Computed from the committed evidence records by a committed script. (a) The risk-managed agent posts the smallest worst-window drawdown of any trading agent on seven of nine datasets, while both long-only references breach the 20 percent per-regime bound almost everywhere and hourly momentum loses 99.3 percent in one window; the two exceptions, hourly crypto and daily FX, are where the trend filter itself whipsaws. (b) The best zero-skill agent's deflated Sharpe as the trial count grows: near certainty at $N = 1$ on short weekly tracks, under 0.43 everywhere by $N = 50$. Deflation is what separates these curves from the bar.

The result is a negative that sharpens the benchmark's claim. The agent is rank-eligible nowhere, under either reliability verdict, and the manner of its failure is the finding. On weekly US indices it clears the bootstrap null ($p = 0.0005$), the probabilistic Sharpe (0.9998), the process gate, and the per-run drawdown bound, and is refused solely on a deflated Sharpe of 0.30 against the 0.95 bar. On seven of the nine datasets its worst-window drawdown is the smallest of any trading agent (Figure 2a), so the discipline registers where it was designed to. What discipline does not manufacture is a deflation-surviving edge.

The two exceptions are themselves evidence. On hourly crypto the trend filter is whipsawed into an 87 percent drawdown in one window, far worse than buy-and-hold's 49, and on daily FX its 33 percent trails buy-and-hold's 20: the same rules that protect capital on daily and weekly equity bars amplify losses where the trend signal churns. Risk management is frequency-dependent, and a benchmark that scores per window is what makes that visible instead of averaged away.

Together with Section 5.3 this shows the gates discriminate rather than blanket-refuse: unhedged beta fails on reliability, and discipline without alpha fails on deflation, each with the quantity that justifies it on the same row. The eligibility bar is therefore not vacuous against the entrants that should fail it. Whether an agent with genuine edge passes is the one question this field of reference agents cannot answer, and Section 7 states it as the outstanding experiment.

5.6 Robustness under perturbation

The self-audit's adversarial-input attack records its own limit: fragility that no submitted run exercises is invisible to a deterministic kernel. The harness now generates perturbed datasets whose bar-to-bar moves are nudged within the empirical range of the original series, a constraint asserted per bar, so every perturbed path is one the market could have printed. Scoring a field across perturbations reports each agent's spread between best and worst per-run PSR. On weekly US indices the risk-managed agent and buy-and-hold both show spreads under 10^{-3}, and a deliberately fragile synthetic agent, one that latches onto an exact price value, shows more than twice the reference spread, which is the separation the diagnostic exists to produce.

5.7 The falsification leg

A benchmark whose gates are strict can still be wrong in the other direction, by rewarding noise. On every dataset the luck floor's best agent has a lower raw return than the best reference agent, with zero exceptions in nine, and at the default configuration no random agent reaches a DSR above 0.423. The floor behaves as a floor.

The grid's loosest corner is the more instructive one. At $N = 1$, which is no deflation at all, a random agent reaches a DSR of 0.999 on weekly crypto and 0.993 on the rates series. A zero-skill agent, scored without a selection correction on a short weekly track, looks nearly certain to have an edge. At $N = 50$ on the same datasets the same agents score 0.423 and 0.000. Neither is eligible at any N, because pass^k refuses them independently. That gap is the deflation doing the work the benchmark exists to do, measured on the market rather than on a synthetic track, and it is the counterpart to Figure 1b: there the track was fixed and the trials grew; here the trials are fixed and the agent is noise.

5.8 What the evidence supports

Taken together, the three findings support one claim and decline another. The claim: applied to the market itself across four asset classes and four bar sizes, the benchmark does not certify any unhedged long-only strategy as safe to hand capital under either definition of reliability, and it reports the drawdown that justifies the refusal. The luck floor never beats a reference agent, the self-audit demotes all eight attacks, and every number here recomputes byte-identically on three operating systems. The claim it declines: that eligibility is attainable by a real agent. Every entrant in these fields is a reference agent with no risk management. Showing that a hedged or risk-managed agent clears the gate, and that the gate is therefore not vacuous, requires such an agent to compete, and Section 7 lists it as the outstanding experiment.

6 Related work

Skill and luck in finance. The distinction the benchmark operationalizes is old. Applying a false-discovery correction to two decades of mutual funds, Barras et al. [2010] find that roughly three quarters had no genuine alpha and that most apparent winners were false positives; Fama and French [2010] reach the same conclusion by a different route. Bailey et al. [2014] show that a high backtest Sharpe is trivially achievable by trying enough configurations, and Harvey and Liu [2015] turn the observation into a haircut that scales with the number of strategies tried. The deflated Sharpe ratio of Bailey and López de Prado [2014] is the instrument this benchmark uses; the Reality Check of White [2000], the superior-predictive-ability test of Hansen [2005] and the step-down procedure of Romano and Wolf [2005] are the field-wide tests it reports. Arnott et al. [2019] set out the reporting protocol a backtest should follow in the era of machine learning, and Section 5 follows it: every Sharpe is reported beside its number of trials, its track length, and the cost profile it was earned under.

Benchmarks for financial agents. Table 5 positions the benchmark against the boards it is most often compared with, on the axes the principles in Section 2 make load-bearing. A comparison drawn only on one's own axes is advertising, so Table 6 draws the same comparison on the axes the rivals hold: every one of them scores real LLM agents, most present a far richer information environment, StockBench runs single-name equities on a genuinely post-cutoff window with a live board, and the Open FinLLM Leaderboard is hosted under neutral governance. SharpeBench's row in that table is empty, its empty cells are the limitations of Section 7, and the two tables together state the actual trade: statistical and adversarial guarantees were bought at the price of field breadth, and the program of Section 7 is to pay that price back without giving up the guarantees. Each mark in both tables records what the cited paper or its public board states, not an inference. FinBen [Xie et al., 2024] ranks GPT-4 first on its trading task with a reported Sharpe of 1.51 ± 1.08, and reports buy-and-hold on the same task at 0.02 ± 0.87; the two intervals overlap, so the board cannot separate its leader from the baseline it is meant to beat. StockBench [Chen et al., 2025] evaluates on a single four-month post-cutoff window and ranks on cumulative return and drawdown, with three seeds per model and no deflation. QuantBench [Wang et al., 2025] names overfitting as an open problem and ranks pipeline performance with no luck-corrected metric. The FINOS Open Financial LLM Leaderboard [FINOS Foundation, 2025] measures financial knowledge, scored on accuracy and F1, and has no trading-performance axis. A 2026 evidence map of LLM-trading research [Xia et al., 2026] coded its nineteen primary empirical studies for reproducibility and found that none reached the top tier. The re-scoring experiment this comparison invites, taking a rival board's published field

Table 5: Positioning along the axes this benchmark adds. A mark means the cited paper states the property of itself; empty means it does not claim it. Read together with Table 6: a comparison drawn only on one's own axes is advertising.

	Deflates	pass^k	Process gate	Costs	Deterministic	Forward commit
FinBen [Xie et al., 2024]						
StockBench [Chen et al., 2025]						
QuantBench [Wang et al., 2025]						
Open FinLLM [FINOS Foundation, 2025]						
τ-bench [Yao et al., 2024]		✓				
SharpeBench (this paper)	✓	✓	✓	✓	✓	✓

Table 6: The same comparison on the axes the rival benchmarks hold. A mark means the cited paper or its public board states the property; empty means it does not. SharpeBench's empty cells are its limitations (Section 7), and three of them, the missing LLM field, the thin information environment, and the absent live board, are the price paid for the guarantees in Table 5.

	LLM agents	Rich info	Single names	Live board	Post-cutoff run
FinBen [Xie et al., 2024]	✓	✓			
StockBench [Chen et al., 2025]	✓	✓	✓	✓	✓
QuantBench [Wang et al., 2025]	✓				
Open FinLLM [FINOS Foundation, 2025]	✓	✓		✓	
τ-bench [Yao et al., 2024]	✓				
SharpeBench (this paper)					

and ranking it under these gates, is now one command: an import adapter converts per-period return series into scoreable submissions, embedding the caveat that imported fields carry no audit trace and unknown trial counts understate deflation, so any demotion shown is a lower bound. The investigation it forced is itself a finding: StockBench publishes environment inputs and summary statistics but no per-agent return series, so the experiment awaits either the authors' raw artifacts or a reproduction with their harness.

Evaluation as a scientific object. The reliability-versus-capability split that motivates pass^k originates in Yao et al. [2024], which asks whether an agent succeeds on all k attempts rather than on at least one. It is now a theme across agent evaluation: Miller [2024] argues for error bars on every evaluation score, Kapoor et al. [2024] for cost-controlled evaluation and adequate holdout sets, and Rabanser et al. [2026] treats reliability as an axis separate from capability and finds that gains in the latter have not bought the former. The surveys that shape this paper's integrity section, Reuel et al. [2024] and Bean et al. [2025], find that most benchmarks neither report the statistical significance of their rankings nor allow their results to be replicated, and recommend an operational construct-validity checklist, which Section B answers. A 2026 survey of financial multi-agent evaluation [Nguyen and Pham, 2026] catalogues five recurring failure modes: look-ahead, survivorship, backtest overfitting, transaction-cost neglect and regime-shift blindness. Deflation answers one. The benchmark's legs are arranged so that each closes a different one, and Section 5 reports that the regime leg is the one that bites.

7 Limitations

No alpha-bearing agent has competed. Every entrant in Section 5 is a reference agent, and the strongest of them, the risk-managed agent of Section 5.5, carries discipline but no edge. The evidence now shows the gates discriminating, refusing beta on reliability and discipline on deflation, each with the quantity that justifies it, so eligibility is not vacuous against entrants that should fail. What it still does not show is an agent being admitted. The experiment that settles it, an agent with genuine edge competing under forward attestation, has not been run, and it is listed first on purpose.

Coverage and observation. Single-name equities are absent for want of a keyless feed, gold because the public daily series were withdrawn, and two of nine datasets fail the realism battery and ship with the verdict recorded. The agent sees twenty trailing closes; the protocol carries fields for fundamentals and news and the simulator leaves them empty, so this is a thinner information environment than FinBen or StockBench present, and the re-scoring experiment now has its adapter but not its data: no rival board publishes per-agent return series, so it awaits their artifacts or a reproduction with their harness.

The reliability gate is strict by construction. The default certifies profitability in every regime, and Sections 5.3 and 5.4 show that both it and a per-run drawdown bound refuse every long-only agent on every dataset containing a downturn. Whether either is the right definition of safe for a given mandate is the operator's choice, now exposed as configuration rather than hidden as a constant; the paper does not claim to have found the right default. On short windows the per-run PSR bar is weak because the statistic scales with $\sqrt{n-1}$, which is a property of the PSR and not a choice.

The measured dispersion is a field statistic. When the field is large enough, σ_{trials} is measured across the submitted agents. A field of reference agents that all lose money has a small dispersion, which lowers the bar, and a field can in principle be assembled to move it. The value is reported on every score so a reader can see it.

Built and not yet public. The arena now drives the full forward lifecycle and its tests publish and verify chained boards, but no window has run against a real future date: there is no hosted intake, no scheduler, and no live board, so the league exists as infrastructure an operator can drive rather than as a running service. External agents are sandboxed by container isolation only; multi-tenant hosting of untrusted submissions remains unbuilt. The perturbation diagnostic of Section 5.6 narrows the self-audit's blind spot but samples perturbations rather than searching adversarially, so a fragility that survives sampling can still hide. The calibration module's epistemic leg reads agreement among an agent's confidence streams as low uncertainty, so unanimous-but-wrong signals read as agreement and the leg is informative only when it reads high.

What changed during this paper. Two of the four findings are corrections to the benchmark as it shipped, a deflation prior in the wrong units and a documentation claim that field-wide significance tests gated rank, and a third correction, that the HMAC chain was not publicly verifiable, is in Section 4. They are reported as findings rather than omitted because a benchmark paper that hides what its own evidence found is the thing the benchmark exists to prevent.

8 Reproducibility statement

The benchmark is a Rust workspace of twelve crates under the MIT and Apache-2.0 licenses, published to crates.io, npm and PyPI at the version stated in Section A, with a pinned toolchain and a lockfile. The scoring kernel forbids unsafe code. Every dataset in Table 7 is committed with a SHA-256 sidecar and a fetch script whose check mode reproduces the committed bytes from the public source. Every figure in this paper is produced by a committed script from the kernel's own formulas, and every table is produced by the commands in Section A from the committed data. Golden fixtures pin the full ranked output of two fields at full floating-point precision and are checked on Linux, macOS and Windows on every commit; a parity test asserts the WebAssembly build and the native kernel produce byte-identical scores. The self-audit runs on every commit and the build fails if any of its eight attacks is not demoted. Continuous integration for the commit this paper was built from is green on all twelve jobs.

No large language model is a component of the benchmark or of any result in this paper. The kernel has no judge. Language models were used to draft and edit prose; no citation in the bibliography was generated by one, and every arXiv entry was checked against its abstract page.

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A Commands that produce every number

All commands run from the repository root at the tagged version. The kernel is deterministic, so each command produces the same bytes on any platform.

Version. Findings 1 and 2 were first observed at v0.2.1. The units fix is v0.3.0. The configurable reliability gate, the per-run drawdown bound, and the evidence sweep that scores both verdicts on identical trajectories are in the release after it. Table 3 was computed against v0.2.1; every other table and figure against the later kernel.

The shipped demonstration (Figure 1a).

```
sharpebench score suites/example_submissions.json
```

The deflation curve (Figure 1b) and the demotion chart.

```
python paper/src/make-figures.py
```

The script reimplements the kernel's PSR, expected-maximum-Sharpe and normal-quantile functions in Python and states that it does; the figures are the kernel's formulas, not an independent computation.

The self-audit (Table 2).

```
sharpebench audit
```

Data realism (Table 7).

```
sharpebench realism --data data/<dataset>.csv
```

Regenerating any dataset and checking its hash.

```
python scripts/data/fetch_crypto.py --interval 1h --check
python scripts/data/fetch_fred.py fx --check
python scripts/data/derive_weekly.py --check
```

The evidence sweep (Table 4, Section 5.2, Section 5.3, Section 5.4, Section 5.7).

```
cargo run --release -p sharpebench-harness \
  --example evidence_sweep -- out.jsonl [dataset] [dsr_bar]
```

One JSON record per (dataset, configuration, agent), 512 per dataset. The grid, seeds, window rule, cost profile and both reliability configurations are constants in the source and are emitted on every record. The optional arguments restrict a run to one dataset or one deflation bar so the grid can be split across processes; because every seed is pinned, the concatenation of parts is identical to a serial run. The analysis that turns the records into the tables is a short script committed beside them under `paper/evidence/`.

The risk-managed agent (Section 5.5) and perturbation report (Section 5.6).

```
cargo run --release -p sharpebench-harness \
  --example risk_managed_eval -- out.jsonl
```

Runs the risk-managed agent beside buy-and-hold and the luck floor on all nine datasets with the documented defaults, both reliability verdicts per row, and appends the perturbation spread report for weekly US indices.

The evidence figures (Figure 2a, Figure 2b).

```
python paper/src/make-evidence-figures.py
```

Every bar and point is a reduction over the committed records; no number is typed into the script.

The forward arena (Section 4).

```
sharpebench arena init <dir>
sharpebench arena open <dir> <window> <deadline> <reveal> [--config c.json]
sharpebench arena commit <dir> <window> <agent> <digest> <salt>
sharpebench arena advance <dir> <epoch>
sharpebench arena score <dir> <window> <data.csv> <reveals.json>
sharpebench arena publish <dir> <window> <key>
sharpebench arena verify <dir> [--pubkey <hex>]
```

Importing a rival field.

```
sharpebench import csv <dir-or-file> --out subs.json [--trials N]
sharpebench score subs.json
```

Golden fixtures and the parity test.

```
cargo test -p sharpebench-core --test golden_scores
cargo test -p sharpebench-sim  --test golden_input
cargo test -p sharpebench-wasm --test native_parity
```

Regeneration requires `SHARPEBENCH_UPDATE_GOLDEN=1` and refuses to run when `CI` is set.

Signing and verifying a board.

```
sharpebench sign   subs.json <hmac-key> board.json --ed25519 <secret>
sharpebench verify board.json --public
sharpebench verify board.json --pubkey <hex>
```

The second form verifies from the document alone. The third pins the key and rejects a board re-signed under a different one.

B Construct validity checklist

Bean et al. [2025] review 445 benchmarks and recommend that a benchmark paper answer an operational checklist. The answers below point to the section that supports each one, and say plainly where the answer is no.

Define the phenomenon. Yes. The phenomenon is skill that survives four corrections: deflation for the number of trials, reliability across seeds and windows, process discipline, and a bootstrap null. Section 2 states it and Equation (3) operationalizes it exactly.

Measure only the phenomenon. Partly. The eligibility predicate measures the phenomenon and nothing else. The scorer also reports fourteen statistics that do not gate, and Section 3.2 separates the two so a reader can tell which number moved the rank.

Construct representative datasets. Partly. Nine datasets across four asset classes and four bar sizes, each scored for realism before use (Section C.2). Single-name equities are absent, two datasets fail the realism battery and ship with the verdict recorded, and the observation is twenty trailing closes (Section 7).

Acknowledge limitations of reused datasets. Yes. Every dataset's source, license, caveats and failing stylized facts are in Table 7 and the data scripts. The commodities file keeps a negative oil close as published and documents the opt-out.

Prepare for contamination. Yes. Sealed held-out datasets under a published hash, a canary token, and a masked view that defeats ticker memorization (Section C.4). Forward commitment before the data exists (Section 4). The forward league that would exercise it has not run.

Use statistical methods. Yes. Every rank is gated on a deflated Sharpe with its number of trials, a block bootstrap with a fixed seed, and a per-run reliability test; field-wide Reality Check, SPA and step-down are reported; a bootstrap confidence interval on the DSR marks tied entries as tied. The annualization convention is stated once in Section 3.1 because getting it wrong was the first finding.

Conduct error analysis. Yes, and it is the paper. Section 5 traces every refusal to the gate that produced it and the quantity that justified it, and two of the three findings are errors in the benchmark itself.

Justify construct validity. Partly, and the gap is named. The benchmark refuses every reference agent for reasons the data makes explicit, and the luck floor never beats a reference agent. What it has not shown is a real agent being admitted, so the claim that eligibility is attainable and the construct is not vacuous awaits the competing agent listed first in Section 7.

Statistical reporting. Every Sharpe in this paper is per period unless stated annualized, and Table 3 gives the conversion at every bar size. Every deflated Sharpe is reported beside its N. All runs use eight execution seeds and six windows; the per-cell record carries both. Costs are the typical profile of Section C. Block bootstrap, not i.i.d., throughout.

Licenses. The benchmark is MIT or Apache-2.0. FRED series are public domain. Binance klines are fetched from the public endpoint under its terms and redistributed as derived daily, 4-hour, hourly and weekly closes. No dataset requires a key.

Maintenance. The benchmark is versioned on crates.io, npm and PyPI with a pinned toolchain; a tag on the main branch publishes to every registry and a verification job confirms each serves the version. Datasets are frozen by hash and regenerated only by a deliberate script run. Golden fixtures change only by a local command that refuses to run in continuous integration. Adding a dataset means adding a hash-pinned file, a fetch script with a check mode, a realism verdict, and a periods-per-year value.

C Simulator, data, replay and contamination

C.1 Simulator and cost model

The data store returns rows at or before the decision index. Execution charges a per-trade fee, seeded slippage, own-order impact that scales with the square root of participation, and financing on gross exposure above one times net asset value. Fills can be capped at a fraction of participation with the remainder deferred. Three named cost profiles ship: none, typical, and worst case. The typical profile is 2 basis points fee, 3 slippage, 50 impact and 5 financing; worst case is 10, 15, 150 and 20 with a participation cap of 10 percent and a two-bar delay. All experiments in this paper use the typical profile. The seeded slippage is what makes different execution seeds produce different runs and gives pass^k something to measure.

C.2 Data

Nine frozen datasets ship with the benchmark, every one keyless and redistributable, each with a SHA-256 sidecar and a deterministic fetch script whose check mode reproduces the committed bytes. Table 7 lists them. Before any agent is scored on a dataset, the dataset is scored: a battery of stylized facts after Cont [2001] tests excess kurtosis, volatility clustering, gain-loss asymmetry, aggregational Gaussianity and time-reversal asymmetry, and issues a verdict. Seven of the nine pass. Weekly US indices fail the aggregation test because 522 bars is too short for it, and FX majors fail time-reversal asymmetry, which is known to be weak in that market. Both ship with the verdict recorded. A realism gate that never fails is not a gate.

Two datasets carry caveats the scripts document. The commodities file keeps the WTI close of -36.98 on 20 April 2020 as published, which dominates its moments; a start date of May 2020 is

Table 7: Frozen datasets. Rows are per symbol. Periods per year is the value every annualized threshold is converted with.

Dataset	Symbols	Bars	Periods/yr	Range	Realism
US indices, daily	SPX DJI IXIC	7540	252	2016–2026	pass
US indices, weekly	same	522	52	2016–2026	fail (n)
Crypto majors, 1h	BTC ETH SOL BNB XRP	24000	8760	2023–2026	pass
Crypto majors, 4h	same	6000	2190	2023–2026	pass
Crypto majors, daily	same	5001	365	2023–2026	pass
Crypto majors, weekly	same	307	52	2020–2026	pass
FX majors, daily	EUR GBP JPY AUD CHF vs USD	4157	252	2010–2026	fail (Zumbach)
Commodities, daily	WTI Brent	4126	252	2010–2026	pass
Rates, daily	10y Treasury yield	4157	252	2010–2026	pass

the documented opt-out. The rates file's close column is a yield in percent, not a price. Single-name equities are absent because no keyless feed exists for them; that is the known gap in coverage.

C.3 Replay and recompute

A run can be captured as a trajectory holding only the agent's raw decisions, the window, and the execution seed, and never any return or self-reported statistic. A separate verifier replays the decisions through the same engine against the frozen dataset and recomputes the score. Capture and replay share one code path, so the round trip is exact by construction, and a test asserts that doubling every order in a captured trajectory recomputes to different returns. An artifact therefore cannot lie about its returns. What it cannot do is prove that the committed strategy produced those decisions; that is the job of the commitment in the next subsection, and the two mechanisms are not yet joined by code.

C.4 Contamination

Two defenses ship. A held-out dataset can be sealed under a published commitment to its hash, so the benchmark can prove later that the data a model was scored on is the data it committed to before scoring. A canary token with the standard training-corpus warning is embedded beside it, so post-hoc detection of a model that trained on the scenarios is possible. Separately, a masked view of any dataset renames symbols and dates to opaque identifiers, which defeats an agent that pattern-matches memorized tickers.

C.5 The units correction, in detail

The fix, shipped in v0.3.0, leaves the per-period kernel untouched and gives the configuration a periods-per-year field so that σ_{trials} is stated annualized and converted once at the point of use. A test asserts the conversion is applied exactly once on the configured path and never on the measured path. Another asserts that an SPX-like series is eligible under an annualized prior of 0.1 at 252 periods per year and ineligible under the same prior at one period per year, which reproduces the old units exactly. A test that the same series clears the original 0.5 prior was not written, because it cannot: 0.5 annualized at 50 trials is a benchmark of 1.14 annualized, and the index sits below it at any track length. That is the correct verdict for that prior, and the benchmark now states it in units a reader can check.